Assignment no. 1

Natural Language Processing
NLP4IL (AIS.ba VZ)
WS 2026

Exercise 1 Build a Polite News Crawler: Implement a domain-specific scraper that collects news articles about Artificial Intelligence. As the base of your crawler use BeautifulSoup.

figureout how we only get news articles about AI

Tasks:

- Crawl at least 3 different news websites.
- Collect at least 200 articles and 100,000 words total.
- Store URL, title, publication date, article text, domain, and crawl timestamp.
- Respect robots.txt.
- Implement rate limiting ($\geq$ 1 second delay).
- Avoid duplicate URLs.

use ars technica and two more interesting ones

Output format (JSONL example):

```
{
  "url":    "...",
  "title":  "...",
  "date":   "...",
  "text":   "...",
  "domain": "..."
}
```

Deliverables:

- Python source code
- Dataset file
- Short reflection discussing ethical considerations, challenges, and data quality issues

Exercise 2 Boilerplate Removal & Text Cleaning. Transform raw HTML files into NLP-ready clean text. Students receive 100 raw HTML files containing navigation, ads, and boilerplate.

Tasks:

- Extract article title and main body text.
- Remove navigation menus, footer, advertisements, scripts, and styles.
- Apply preprocessing: lowercasing, tokenization, stopword removal.
- Compute token count before and after cleaning.
- Calculate percentage of removed content.
- Identify top 20 most frequent words

Deliverables:

- Cleaned dataset (JSON format)
- Jupyter Notebook with analysis
- Short reflection explaining the importance of boilerplate removal and impact on NLP/LLM training

When using Large Language Models (eg. ChatGPT, Copilot, ...) to produce output that is handed in, also state the prompt and model used in the submission. Submissions that are not marked and are noticeably AI generated will be graded with zero points!

Submission: electronically via Moodle; Deadline: Mo, Mar 9, 2026, 23:59.